例題 5.7　2 変数 x, t を含む関数 $u(x,t)$ に対して，偏微分方程式

$$\frac{\partial u(x,t)}{\partial x} = \frac{\partial u(x,t)}{\partial t} + u(x,t) \quad (0 < x < \infty,\ t > 0)$$

を解け。ここで，$u(x,t)$ は $t = 0$ のとき初期値を $u(x,0) = e^{-2x}$，$\lim_{x\to\infty} u(x,t) = 0$ とする。

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P102. 例5.7.

$$\frac{\partial u(x,t)}{\partial x} = \frac{\partial u(x,t)}{\partial t} + u(x,t).$$

$\downarrow$ convolution

$$\mathcal{L}\left[\frac{\partial u(x,t)}{\partial x}\right] = \mathcal{L}\left[\frac{\partial u(x,t)}{\partial t}\right] + \mathcal{L}[u(x,t)]$$

$$\frac{\partial}{\partial x} U(x,S) = sU(x,S) - u(x,0) + U(x,S).$$

$$\frac{dU}{dx} - (S+1)U = -e^{-2x}$$

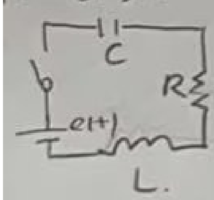
$$U(x,S) = \frac{1}{S+3} e^{-2x} + Ce^{(S+1)x}.$$

$\because \lim_{x\to\infty} u(x,t) = 0.$

$\therefore U(x,S) = \frac{1}{S+3} e^{-2x}.$

$u(x,t) = e^{-3t} e^{-2x}$

5.3.1.



$$e(t) = L\frac{di(t)}{dt} + Ri(t) + \frac{1}{C}\int i(t)\,dt.$$

$\downarrow$ convolution

$$\frac{E}{S} = L\mathcal{L}\left[\frac{di(t)}{dt}\right] + R\mathcal{L}[i(t)] + \frac{1}{C}\mathcal{L}[q(t)]$$

$$= L\{s^2Q(S) - Sq(0) - i(0)\} + R\{SQ(S) - q(0)\} + \frac{1}{C}Q(S).$$

$\downarrow$ $q(0)=0$ $i(0)=0$.

$$\frac{E}{S} = LS^2Q(s) + RSQ(s) + \frac{1}{C}Q(s).$$

$$Q(s) = \frac{E}{S(LS^2+RS+\frac{1}{C})} \rightarrow I(s) = \frac{E}{LS^2+RS+\frac{1}{C}} = \frac{E}{L}\cdot\frac{1}{S^2+\frac{R}{L}S+\frac{1}{CL}}$$

$$2a = \frac{R}{L},\ b = \frac{1}{\sqrt{CL}} \qquad = \frac{E}{L}\frac{1}{(S+\frac{R}{2L})^2+\frac{1}{CL}-\frac{R^2}{L^2}}$$

① $a<b$.

$$i(t) = \mathcal{L}^{-1}[I(S)] = \frac{E}{L}\frac{e^{-at}}{\sqrt{b^2-a^2}}\sin\sqrt{b^2-a^2}\,t$$

② $a>b$.

$$i(t) = \frac{E}{L}\frac{e^{-at}}{\sqrt{a^2-b^2}}\sinh\sqrt{a^2-b^2}\,t$$

③ $a=b$.

$$i(t) = \frac{E}{L}te^{-at}$$